THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Understanding the Weather** | Class 7 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 14 degrees C	34 - 20
2	**(b)** thermometer and hygrometer	Dew point
3	**(a)** cloudy, humid or coastal	Small range
4	**(b)** a smaller range	Moderation by the sea
5	**(a)** clouds trap outgoing heat	Insulation
6	**(a)** 13 degrees C	6.5 x 2
7	**(c)** a storm is approaching	Pressure falls before storms
8	**(c)** from sea to land	High to low
9	**(a)** Both A and R are true and R is the correct explanation of A.	R gives the reason for the direction of wind in A.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but R is unrelated to the calculation of temperature.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Show climate trends (1); Reveal changes (1).	1+1
12	Latitude and surface (1); Different heating of land and sea (1).	1+1
13	Convective storms (1); Rapid changes (1).	1+1
14	Large pressure difference (1); Air moves faster (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Mean 30, 32, 28 (1); Range 12 each (1); Working shown (1).	1+1+1
16	Low pressure centre (1); Strong winds (1); Possible storm (1).	1+1+1
17	Evacuation (1); Fishermen warnings (1); Property protection (1).	1+1+1
	OR: High humidity slows sweat evaporation (1); Feels hotter (1); Low humidity feels drier (1).	1+1+1
18	Cloud tracking (1); Early warnings (1); Global data (1).	1+1+1
	OR: Chaos (1); Data (1); Models (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) (38 + 24) / 2 = 31 degrees C (1). (ii) The amount (depth) of rainfall (1). (iii) A thermometer (1). (iv) The mean is worked out from the two extremes (1); long records help to understand patterns and make forecasts (1).	1+1+1+2
20	Heating (1); Low pressure (1); Wind (1); Humidity (1); Rain (1).	1+1+1+1+1
	OR: History (1); Observation network (1); Forecasts (1); Cyclone warnings (1); Benefits (1).	1+1+1+1+1

Mistakes to avoid:

- Mixing up the instruments: **thermometer** - temperature, **rain gauge** - rainfall, **barometer** - pressure, **wind vane** - direction.
- Using the wrong formula for mean daily temperature; it is **(maximum + minimum) / 2**, not the difference.
- Writing that wind blows from low to high pressure; it blows from **high to low pressure**.
- Confusing weather with climate; weather is the **day-to-day** state of the atmosphere, climate is the long-term pattern.
- Forgetting the units: rainfall in **mm**, temperature in **degrees C**, pressure in **millibars**.
- Stating that clothes dry faster in high humidity; they dry faster when humidity is **low**.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Understanding the Weather** | Class 7 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** urban heat island and climate change	Heat retention
2	**(c)** early warnings and evacuation	Preparedness
3	**(b)** cloud or fog formation	Saturation
4	**(a)** to avoid trees and buildings blocking rain	Accurate measurement
5	**(b)** evaporates water and moves the air	Solar energy
6	**(c)** dry summer	Dry spell
7	**(c)** to measure air temperature without direct sunlight	Accuracy
8	**(d)** rises	Air can hold less vapour
9	**(c)** A is true but R is false.	A is true; R is false, a barometer measures pressure and a rain gauge measures rainfall.
10	**(d)** A is false but R is true.	A is false, clothes dry faster when humidity is low; R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Predicts storms (1); Safety (1).	1+1
12	Plan travel (1); Stay safe (1).	1+1
13	Air cools (1); Vapour remains (1).	1+1
14	Traps night heat (1); Blocks daytime sun (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Altitude (1); Temperature (1); Freezing (1).	1+1+1
16	Transpiration (1); Shade (1); Cooling (1).	1+1+1
17	Block sun in day (1); Trap heat at night (1); Smaller range (1).	1+1+1
	OR: Based on observation (1); Not always reliable (1); Combine with science (1).	1+1+1
18	Heat island (1); Less evaporation (1); Rain run-off (1).	1+1+1
	OR: Learning (1); Local data (1); Interest (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) From the sea (high pressure) (1). (ii) Over the land (1). (iii) A wind sock (1). (iv) Wind moves from high-pressure to low-pressure areas (1); the warm land has low pressure, so cooler air from the sea moves in to fill it (1).	1+1+1+2
20	Mean max (1); Mean min (1); Range (1); Pattern (1); Interpretation (1).	1+1+1+1+1
	OR: Sowing (1); Irrigation (1); Harvest (1); Crop insurance (1); Forecasts (1).	1+1+1+1+1

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- Using the wrong formula for mean daily temperature; it is **(maximum + minimum) / 2**, not the difference.
- Writing that wind blows from low to high pressure; it blows from **high to low pressure**.
- Confusing weather with climate; weather is the **day-to-day** state of the atmosphere, climate is the long-term pattern.
- Forgetting the units: rainfall in **mm**, temperature in **degrees C**, pressure in **millibars**.
- Stating that clothes dry faster in high humidity; they dry faster when humidity is **low**.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Understanding the Weather** | Class 7 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** places with equal pressure	Pressure lines
2	**(a)** A received more rain	Compare depth
3	**(a)** weather varies daily while climate is a long-term pattern	Concept
4	**(c)** stronger the winds	Greater pressure gradient
5	**(a)** drying clothes outdoors	Little evaporation
6	**(a)** moist winds are forced to rise and cool	Relief rain
7	**(c)** seasonal trends	Long-range limits
8	**(d)** 37 and 31	Mean 34
9	**(a)** Both A and R are true and R is the correct explanation of A.	R follows from the definition of humidity in A and explains drying.
10	**(d)** A is false but R is true.	A is false, a wind vane shows direction, not speed; R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Run-off (1); No absorption (1).	1+1
12	No warm sea (1); Friction (1).	1+1
13	Contains most vapour (1); Convection occurs (1).	1+1
14	Dry clear air (1); Rapid heating and cooling (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Drought (1); Flood (1); Crop choice (1).	1+1+1
16	Total = 772 mm (1); Wettest = fifth month (1); Monsoon pattern (1).	1+1+1
17	Low pressure rising air (1); Cloud formation (1); Rain (1).	1+1+1
	OR: Land heats faster (1); Low pressure on land (1); Cool air from sea (1).	1+1+1
18	Temperature (1); Humidity (1); Together for comfort (1).	1+1+1
	OR: Latitude: sun's angle (1); Altitude: thinner air (1); Combined effect (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) The amount of water vapour in the air (1). (ii) In low humidity (1). (iii) The India Meteorological Department (1). (iv) Air that is nearly full of vapour can take little more (1); so water from the clothes evaporates slowly (1).	1+1+1+2
20	Meaning (1); Time scale (1); Examples (1); Importance for farming (1); Importance for planning (1).	1+1+1+1+1
	OR: Instruments (1); Location (1); Readings (1); Records (1); Analysis (1).	1+1+1+1+1

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- Writing that wind blows from low to high pressure; it blows from **high to low pressure**.
- Confusing weather with climate; weather is the **day-to-day** state of the atmosphere, climate is the long-term pattern.
- Forgetting the units: rainfall in **mm**, temperature in **degrees C**, pressure in **millibars**.
- Stating that clothes dry faster in high humidity; they dry faster when humidity is **low**.

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